

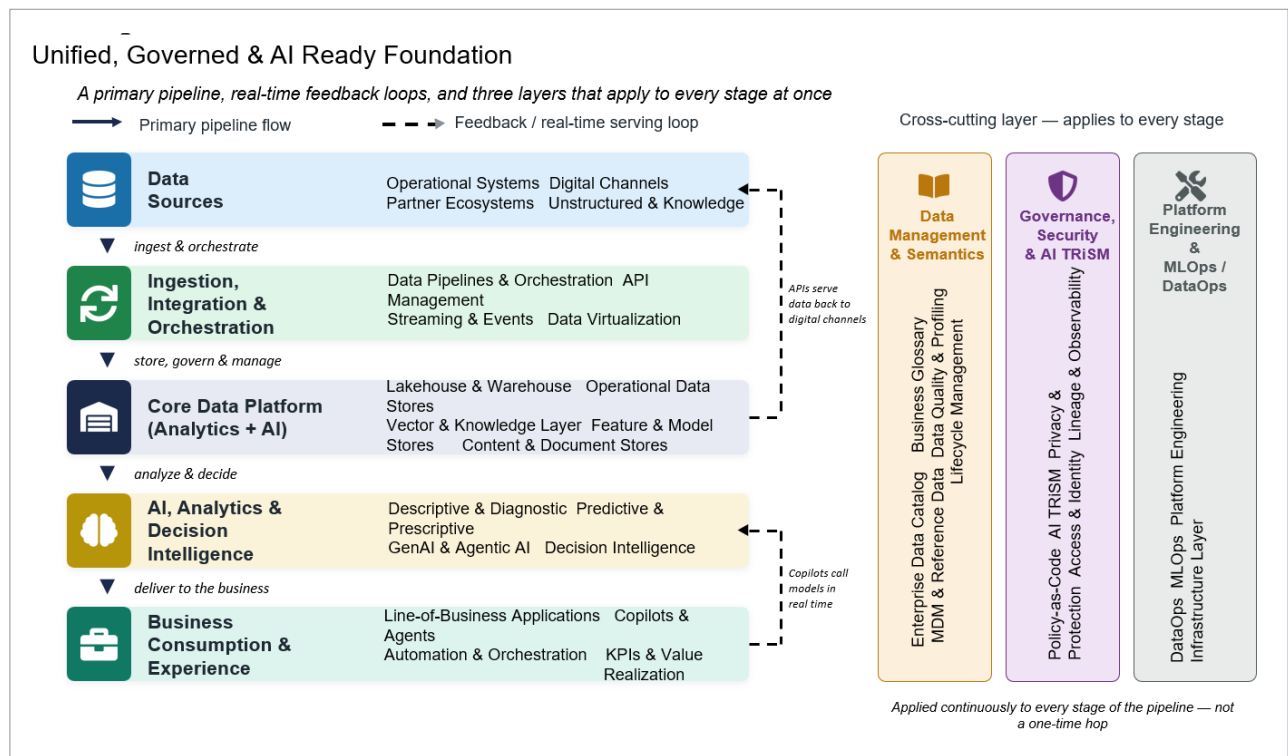


Figure 1: Enterprise data architecture for AI

- **Operational systems:** These are the systems that run the day-to-day business: ERP, CRM, domain platforms, billing, HR, etc. These remain the backbone of structured, transactional data.
  - **Digital channels:** The web, mobile, API, and customer-portal through which customers and employees interact directly with the enterprise are included in this category. These channels are not purely a source; they also receive personalised or real-time data back through APIs.
  - **Partner ecosystems:** These are the B2B integrations, data exchanges, and marketplaces that bring external, third-party data into the enterprise's view.
  - **Unstructured and knowledge data:** This category includes documents, emails, videos, knowledge bases, and ontologies. It has grown in strategic importance because it is precisely the content that large language models and retrieval-augmented generation (RAG) pipelines depend on.
- Ingestion, integration and orchestration:** This layer helps to move data from source into the platform reliably, securely, and in the right cadence like batch, streaming, or on-demand.
- **Data pipelines and orchestration:** Comprises engines that sequence and monitor data movement, paired with pipeline observability so failures and delays are visible before they become business problems.
  - **API management:** Includes gateways, throttling, versioning, and security policy enforcement for every API-based integration, ensuring that data movement through APIs is controlled rather than *ad hoc*.
  - **Streaming and events:** These are event hubs and pub/sub infrastructure (e.g., Kafka-style platforms) that support event-driven integration where near-real-time movement is required.
- Data virtualisation:** This lets consumers query across multiple heterogeneous stores without first physically consolidating the data, reducing duplication and latency for enterprise usage.
- Core data platform (analytics + AI):** This is the heart of the architecture that acts as an AI-ready layer. This unified storage and serving layer is the place where data lives and is made available for both traditional analytics and AI workloads from a single, governed foundation.
- **Lakehouse and warehouse:** This category combines the flexibility of a data lake with the performance and semantic structure of a warehouse, including reusable semantic models that give consistent business meaning to raw tables.
  - **Operational data stores (ODS):** Supports near-real-time reporting for use cases that cannot wait for a batch cycle.